

Prem Babu Kanaparthi

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Summary

AI Engineer building Python LLM, RAG, agent, and classical ML systems. Shipped routing APIs, eval gates, experiments, and AI products with latency, cost, and reliability gains.

Skills

- AI and Machine Learning: LLMs, Generative AI, Agentic AI, RAG, Prompt Engineering, PyTorch, Scikit-learn, XGBoost
- Evaluation and Experimentation: LLM Evaluation, Eval Harnesses, Retrieval Evaluation, Drift Monitoring, Responsible AI
- Deployment and Cloud: Python, FastAPI, LiteLLM, AWS Bedrock, SageMaker, Model Serving, Docker, CloudWatch, MCP
- Data and Retrieval: SQL, SQLite, PostgreSQL, FAISS, BM25, Cross-Encoder Reranking, Structured Data, Reproducible Pipelines

Experience

Rochester Institute of Technology Jan 2026 - Apr 2026
Graduate Researcher . Rochester, NY

- Extended Stanford's Generative Agents with a 72-feature LSTM predictive model, controlled scenarios, ablations, and statistical comparisons across 205,940 agent-step records.
- Validated statistically significant emergent emotion patterns against Gallup 2024 human baselines and documented model behavior, limitations, and failure modes (poster).

Concentrix + Webhelp Feb 2024 - Jul 2024
Generative AI Engineer . Newark, CA

- Owned the production LLM inference routing layer for 50K daily requests across 3 foundation models on AWS Bedrock and SageMaker; used LiteLLM, provider fallback, and latency-aware routing to cut p95 latency from 4s to 1.5s.
- Built production LLM evaluation and safety gates with CloudWatch logging, hallucination checks, drift monitoring, and a 500-case test set; reduced incidents by 42% and MTTD by 35%.
- Reduced monthly inference spend from \$45K to \$37K through cost-aware routing, prompt caching, and provider fallback while maintaining 95%+ task success.

AlphaBits Technologies Aug 2023 - Jan 2024
Data Science Intern . Bengaluru, India

- Rebuilt a Python search-ranking experimentation pipeline with preprocessing, feature generation, SQL-backed analysis, and 5 model variants; cut iteration time by 90% and improved relevance by 10%.
- Created an offline ranking evaluation harness tracking relevance, error cases, and preprocessing differences across model variants before deployment.

iNeuron AI Jun 2023 - Aug 2023
ML Engineer Intern . Bengaluru, India

- Developed XGBoost classifiers for learner dropout risk (0.86 AUC across 12,000 learners) and phishing URL detection (20+ URL/domain features, 92% accuracy).
- Built a behavioral feature pipeline converting raw activity logs into 24 rolling-window features with drift and quality checks for retraining.

Education

Rochester Institute of Technology Aug 2024 - May 2026
Master of Science in Artificial Intelligence, completed May 2026 . Rochester, NY
National Institute of Technology Silchar Aug 2020 - May 2024
Bachelor of Technology in Computer Science, completed May 2024 . Silchar, India

Projects

QuanTRAG - Grounded Financial Research Copilot GitHub

- Built cited retrieval over SEC filings with 4 retrieval modes (dense, sparse, hybrid, and reranked), hallucination checks, abstention, metadata filtering, and a 52-question benchmark.

PackAI - Multi-Agent Coding Orchestration GitHub

- Built an agentic coding workflow with agent selection, dependency-ordered DAG planning, multi-file coordination, dashboard controls, and reusable task playbooks; completed 68% of previously failed benchmark tasks.

Nori - AI Job Discovery and Automation Platform norinote.xyz

- Built a 0-to-1 platform with bounded async scrapers, ATS adapters, Redis queues, sentence-transformer ranking, and scheduled workflows backed by a 31,000+ company ATS registry; surfaced 120+ roles/day.